Multiphysics simulation of parametric effects on IPMC actuation dynamics and back-relaxation

OPEN ACCESS
IOP Publishing

Smart Materials and Structures

Smart Mater. Struct. **34** (2025) 095010 (11pp)

https://doi.org/10.1088/1361-665X/adff43

Multiphysics simulation of parametric effects on IPMC actuation dynamics and back-relaxation

Yu-Tung Chen* **and Kamran Behdinan**

Advanced Research Laboratory for Multifunctional Lightweight Structures (ARL-MLS), Department of Mechanical and Industrial Engineering, University of Toronto, Toronto, Canada

E-mail: jasmineytc.chen@mail.utoronto.ca

Received 7 May 2025, revised 13 August 2025
Accepted for publication 26 August 2025
Published 3 September 2025

Abstract

Ionic polymer–metal composites (IPMCs) are a type of smart material capable of large, reversible deformation under low applied voltage. Their flexibility, biocompatibility, and ability to perform underwater make them promising candidates for soft robotics and biomedical devices. However, their application is often limited by their low actuation force and back-relaxation under constant voltage. While many efforts have been made to optimise the performance through material and geometry alteration, a comprehensive investigation of how specific material properties influence the actuation dynamics remains limited. This study attempts to investigate how material parameters and electrical input can influence the actuation behaviour of IPMCs using multiphysics simulation. A 2D finite element model, with consideration of coupled ion–water transport and mechanical deformation, was used to analyse the role of transport (diffusivity, permeability), electrical (dielectric constant), and mechanical (elastic modulus) properties on the actuation performance. Results show that transport-related parameters predominantly affect the transient response, while others influence both transient and steady-state displacement. Specifically, increasing the dielectric constant and diffusion coefficient enhances overall deformation, whereas greater hydraulic permeability and elastic modulus tend to suppress it. Additionally, voltage studies revealed that combining a high AC amplitude with a low DC bias can reduce back-relaxation without compromising actuation performance. These findings clarify the individual roles of material parameters in IPMC deformation dynamics and provide potential voltage modulation strategies to mitigate back-relaxation and improve long-term stability.

Keywords: IPMC, parametric analysis, actuation, FEM, back-relaxation

* Author to whom any correspondence should be addressed.

1. Introduction

Ionic polymer–metal composites (IPMCs) are a type of smart material that deforms under electrical stimuli and, conversely, generates electrical signals under applied load. Due to this reversible nature, they are suitable as both sensors and actuators [1, 2]. IPMCs typically consist of a polymer membrane, such as Nafion or Flemion, with metal electrodes plated on both surfaces. When an electric field is applied, hydrated cations within the polymer migrate toward the cathode, carrying water molecules with them. This causes a redistribution of internal pressure and swelling on one side of the membrane, leading to a macroscopic bending deformation [3–5]. IPMCs are known to be capable of undergoing large, reversible deformations under low applied voltage (typically 1–5 V). Combined with their soft, lightweight structure and ability to function in wet environments, they are considered promising candidates for actuator applications in fields such as biomedical devices, aerospace systems, and soft robotics.

Despite the many advantages, there still exist some challenges that limit their use in certain applications. Among the most critical limitations are their low actuation force, slow response time, and undesirable back-relaxation behaviour under sustained electric fields, which are undesirable qualities for actuators [5, 6]. Specifically, back-relaxation is an important phenomenon that happens in IPMC that needs to be addressed. This is when the actuator slowly returns back to its original position while still under constant electrical load. This loss of displacement over time poses a significant challenge for applications requiring stable and sustained actuation. Therefore, understanding the deformation dynamics and the mechanisms underlying back-relaxation is essential for improving the performance and reliability of IPMC actuators.

Extensive research has been done aimed at improving their performance through material, geometric, and operational modifications. For instance, several studies have investigated how the choice of counter-ion influences the actuation performance of IPMCs [7–9]. These works have shown that ion size, hydration behaviour, and mobility play critical roles in determining the electromechanical response. IPMCs incorporating smaller cations such as Li^+ and H^+ typically exhibit larger blocking forces and faster actuation rates due to their enhanced mobility and stronger hydration effects within the polymer network [7, 8]. Conversely, larger or less hydrated ions tend to reduce actuation speed and force but may offer improved stability and reduced back-relaxation, particularly under ambient conditions with limited humidity. While smaller counterions generally improve the actuation speed, [9] suggests that bulkier ions can achieve greater charge-specific displacement, or ‘pumping’ efficiency, despite their slower actuation speed. These studies demonstrate how the choice of counter-ion influences the water uptake within the membrane, affecting the actuation performance. On a similar note, [10] explored how the water content affects the actuation performance. Their finding shows that higher water content can lead to a more pronounced back relaxation behaviour due to the presence of free water molecules in the membrane. Others have also explored how increasing the membrane thickness and length can improve the blocking force and deformation capacity of IPMCs [11–13]. Experimental results show that output force increases with a thicker membrane, as the greater cross-sectional area and mechanical stiffness enable the material to resist external loads more effectively. However, this comes at the cost of smaller tip deformation and slower response time. This reduction in bending displacement is due to the increased flexural rigidity of the thicker structure, which requires more energy to deform. Additionally, the longer diffusion path for ions and water molecules in thicker membranes slows down the electroosmotic transport process, thereby delaying the actuation response. Some studies also suggest that the water uptake changes with the change in membrane thickness [11, 13], which could be another reason contributing to the effect in actuation performance. Lastly, to improve the mechanical properties of polymers, nanoparticles and fillers are often introduced to improve their mechanical properties [14, 15]. In the context of IPMCs, coating the Nafion membrane with CNT- or graphene-based electrodes has been shown to not only improve its mechanical properties but also increase its water uptake and enhance bending actuation performance [16].

These experimental investigations suggest that altering external factors such as geometry and hydration can significantly influence the performance of IPMC actuators. However, due to the complex and highly coupled nature of ionic transport, electrostatics, and mechanical deformation in IPMCs, modifying one factor often simultaneously affects multiple intrinsic parameters for multi-physics modelling, such as the diffusion coefficient, elastic modulus, and dielectric constant. As a result, it is often difficult to isolate the individual contributions of these parameters through experimental methods alone, making it challenging to fully investigate the mechanisms governing actuation behaviour.

The objective of this study is twofold. First, a parametric study is conducted to investigate how key material and transport parameters, specifically dielectric constant, diffusion coefficient, hydraulic permeability, and elastic modulus, influence the actuation dynamics and back-relaxation behaviour of IPMC actuators. Second, a voltage study explores how different types of electrical input, including DC, AC, and combined AC+DC voltages, affect actuation performance and relaxation characteristics. Both studies are carried out using a multiphysics-based finite element method (FEM) that couples ion–water transport with mechanical deformation. Unlike previous work that focused primarily on experimental optimisation or single-parameter effects, this study provides a comprehensive simulation-based analysis. In particular, the voltage study proposes a modulation strategy using biased AC signals to mitigate back-relaxation, offering new insights for improving the long-term performance and stability of IPMC actuators.

2. Methods

The parametric study and voltage study are conducted through numerical simulations using a two-dimensional FEM implemented in COMSOL Multiphysics. In both studies, the analysis focuses on the time-dependent displacement of the IPMC actuator to evaluate how variations in specific parameters influence the deformation dynamics. Key performance indicators include peak displacement (representing the transient response), steady-state displacement, and back-relaxation behaviour. Displacement is measured at a point 15 mm from the fixed boundary to capture the tip deformation of the actuator. Unless otherwise specified, simulations are carried out under an applied voltage of 2 V with a total duration of 50 s to ensure that the system reaches steady-state. This section provides an overview of the mathematical model used to describe the coupled multi-physics of IPMC actuation, as well as the details of the simulation setup used in the finite element implementation.

2.1. Multiphysics modelling equations

The actuation mechanism of IPMCs, as introduced earlier, involves the migration of cations and associated water molecules under an applied electric field, resulting in localised swelling and bending deformation. Based on this mechanism, the deformation of water-based IPMCs can be described as two sequential and coupled processes: (1) electrically induced mass transport of ions and water and (2) mechanical deformation driven by the resulting redistribution of mass within the membrane.

In this study, the multiphysics model proposed by Zhu *et al* [17] is adopted, which accounts for both the transport of cations and water molecules and couples with the mechanical deformation of the IPMC membrane. Many existing models focus primarily on cation transport [18–20], and the deformation attributed to swelling effects is sometimes represented through a fictitious body force [19, 20]. In contrast, Zhu's model offers a more physically grounded approach by governing mechanical response through volumetric strain that is directly related to local water concentration. It also incorporates internal eigenstresses, osmotic and electrostatic stresses, that arise from spatial concentration gradients. While the model assumes ideal electrode behaviour and neglects chemical reactions such as water electrolysis, it has been shown to be capable of capturing the deformation dynamics, including back-relaxation, given that the effect of water loss and electrolysis is not significant. The key advantage of using this framework lies in its ability to represent the interplay between ion–water transport and stress generation. To enhance numerical stability while retaining essential physical mechanisms, this study takes advantage of the simplified version of Zhu's formulation by applying an extended eigenstress approach that leverages the opposing effects of osmotic and electrostatic forces.

This section gives a summary of the mathematical model from Zhu *et al* [17] and is presented in two sections: the first describes the equations governing mass transport, focusing on ion and water fluxes; the second section details the relationships for stress and strain, highlighting how the redistributed mass within the membrane translates into mechanical deformation.

2.1.1. Ion and water molecules transport.

The transport of cations and water molecules is modelled using the general Nernst–Planck equation (equations (2) and (3)), coupled with Poisson's equation (equation (1)) to describe the distribution of electric potential and charge across the membrane. Unlike other models that also employ the Nernst–Planck framework, this formulation includes a pressure diffusion term and accounts for the coupled transport behaviour between cations and water molecules, allowing for a more accurate representation of mass redistribution under electrochemical loading [17],

$$\nabla^2\phi = -\frac{\rho}{\varepsilon} = -\frac{F\left(c_I - c^-\right)}{\varepsilon} \tag{1}$$

$$J_I = -d_{II}\left(\nabla c_I + \frac{Z_I F c_I}{RT}\nabla\phi\right) - N_{dI} d_{WW}\nabla c_W - c_I K\nabla p \tag{2}$$

$$J_W = -d_{WW}\nabla c_W - N_{dW} d_{II}\left(\nabla c_I + \frac{Z_I F c_I}{RT}\nabla\phi\right) - c_W K\nabla p \tag{3}$$

where ϕ is the potential, ρ is the charge density, ε is the effective permittivity, J_i is the flux density of cations and water molecules in the polymer. c_i is the concentration of the cation and water molecules. p is the total water potential, F is the Faraday constant, R is the gas constant, T is the temperature, and K is the hydraulic permeability coefficient [17].

The drag coefficients are introduced to couple the fluxes of cations and water, representing the amount of water carried by each cation and vice versa. To account for regions with high cation concentration, where a single cation can no longer carry as many water molecules as it would in lower concentration regions, a dynamic drag coefficient is introduced. The dynamic drag coefficient, denoted N_{dI} and N_{dW} is described in equations (4) and (5) [17], where n_{dI} and n_{dW} are the static drag coefficients for cations and water, and c_{W0} is the initial water concentration:

$$N_{dI} = n_{dI}\cdot\left(\frac{c_I}{c_W} > \frac{c^-}{c_{W0}}\right) + \frac{c_I}{c_W}\frac{d_{II}}{d_{WW}} n_{dW}\cdot\left(\frac{c_I}{c_W} \leqslant \frac{c^-}{c_{W0}}\right) \tag{4}$$

$$N_{dW} = n_{dW}\cdot\left(\frac{c_I}{c_W} \leqslant \frac{c^-}{c_{W0}}\right) + \frac{c_W}{c_I}\frac{d_{WW}}{d_{II}} n_{dW}\cdot\left(\frac{c_I}{c_W} > \frac{c^-}{c_{W0}}\right). \tag{5}$$

Lastly, the mass continuity equation (equation (6)) is needed to complete the dynamic transport equation for water and cation,

$$\frac{\partial c_i}{\partial t} + \nabla J_i = 0, (i = I, W). \tag{6}$$

2.1.2. Stress strain analysis. A stress analysis is required to determine the pressure within the cluster to account for the pressure effect in the mass transfer equation. The total pressure contains two parts: hydrostatic pressure (p_h) and eigen stresses (σ^*) generated by the redistribution of liquid [17]. Equation (7) defines the total pressure within the cluster,

$$p = p_h + \sigma^*. \tag{7}$$

Hydrostatic pressure:
Hydrostatic pressure arises from the fluid confined within the cluster spaces and is primarily governed by the elastic stress (σ_e) exerted by the polymer on the solid-liquid interface. Equation (8) defines the relationship between the hydrostatic pressure within the cluster and the elastic stress from the polymer,

$$p_h = -\sigma_e. \tag{8}$$

Under small strain conditions, Nafion's high cross-link density allows its mechanical response to be effectively captured using a Neo-Hookean linear elastic model, without accounting for viscoelasticity. Additionally, by assuming isotropic expansion and contraction of the ionic clusters and treating the material as incompressible, the elastic stress can be defined. Assuming no external mechanical stimulation and the deformation is only caused by internal stress, the equation (9) is derived in [17]:

$$-\sigma_e = \frac{E_{\mathrm{dry}}}{3}\left(\left(\frac{B}{B_0}\right)^{-4} - \left(\frac{A}{A_0}\right)^{-4}\right) \tag{9}$$

where E_{dry} is the elastic modulus of dry Nafion, and B and A are the outer and inner radius of the cluster. Since the radius of the cluster is related to the volume change of the water, equation (9) can be defined as a function of the volume fraction of water, w_v, shown in equation (10) [17],

$$p_h = \frac{E_{\mathrm{dry}}}{3}\left(\left(\frac{1+w_v}{1+w_{v0}}\right)^{-4/3} - \left(\frac{w_v}{w_{v0}}\right)^{-4/3}\right). \tag{10}$$

Eigen stresses:
The mass redistribution also generates eigen stresses, including osmotic pressure and electrostatic stresses, which further contribute to the distribution of water and cations. In [17], a simplified eigenstress equation is proposed, shown in equation (11). This is done by taking advantage of the opposite tendency between the osmotic stress and electrostatic stress within the membrane,

$$\sigma^* = -\varphi\frac{RT}{\overline{V}_W}\ln\left(1+\frac{c_I}{c_w}\right) \tag{11}$$

where $\overline{V}_W$ is the partial molar volume of water and φ is the extended osmotic coefficient that ranges from negative to positive. In the case of a negative coefficient, it indicates that the influence of electrostatic stress is more significant and vice versa.

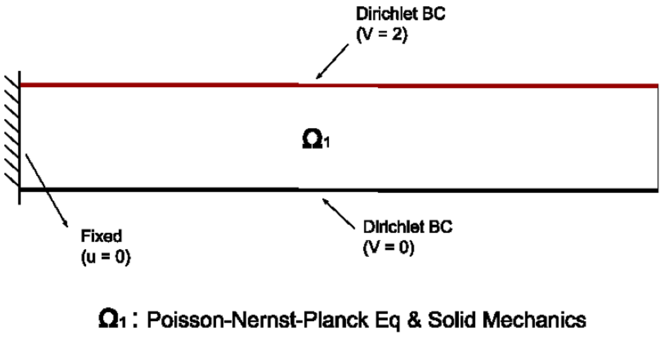

Figure 1. Schematic representation of the domain physics and boundary conditions.

Finally, the deformation of the material is directly driven by changes in water content, which alter the volumetric strain. Given this dependence, the strain (ε_V) can be formulated in terms of the local water volume fraction, as described in equation (12) [17]:

$$\varepsilon_V = \frac{1}{1+w_{v0}}\left(w_v - w_{v0}\right). \tag{12}$$

According to the small strain assumption, the linear eigen strain (ε_{ii}), shown in equation (13) can therefore be obtained:

$$\varepsilon_{ii} = \frac{1}{3}\varepsilon_V \tag{13}$$

2.2. Simulation setup

2.2.1. COMSOL interface. COMSOL Multiphysics is utilised to perform the numerical analysis for the parametric study and voltage study. The Poisson's equation and Coefficient Form PDE interfaces under Mathematics in COMSOL are used to implement Poisson's equation (equation (1)) and the flux equations for cation and water transport (equations (2), (3) and (6)). To model the deformation of the IPMC, the solid mechanics interface with a plane strain assumption is implemented. The internal strain (equation (13)) is induced using the initial stress/strain node. Figure 1 illustrates the domain physics and boundary conditions.

2.2.2. Electrical stimuli. To set up the electrical stimuli, a step function is used to represent a DC voltage input of 2 V for the parametric study. Smoothing is applied to the step function to facilitate numerical convergence. In the voltage study, an AC voltage is introduced. To implement this, a smoothed step function is first used to initiate the simulation, followed by the application of a sinusoidal function starting at 1 s. This gradual transition from zero to the sinusoidal load helps avoid abrupt changes in the electric field, improving stability and ensuring more reliable solver convergence.

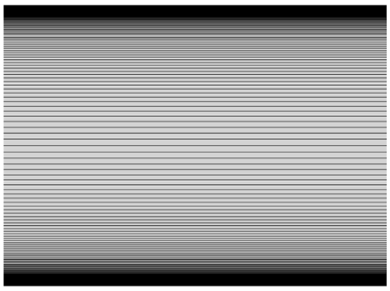

Figure 2. Schematic representation of the mesh.

2.2.3. Mesh. It is reported in literature that the concentration of cations and water molecules mainly changes near the electrodes, whereas the concentration stays almost constant in the bulk of the membrane. Therefore, mapped meshing composed of quadrilateral elements is applied so the mesh is finer near the electrode boundaries and coarser in the middle of the membrane to relieve the computation load. The final mesh consisted of approximately 10 000 elements. Figure 2 shows a schematic representation of the meshing strategy.

2.2.4. Parameters. The deformation behaviour of IPMC actuators can vary significantly depending on fabrication methods and environmental conditions, and in some cases, a back-relaxation effect is observed following the initial actuation. These variations are often reflected in their material and transport properties, such as ionic diffusivity, elastic modulus, and hydraulic permeability. Accordingly, back-relaxation is not the result of a single parameter alone, but rather the outcome of a complex interplay among multiple coupled parameters.

As a result, analysing parameter sensitivity using only one baseline configuration may fail to capture the full picture of actuation behaviour. Therefore, to fully investigate how individual parameters influence the actuation dynamics, two distinct sets of parameter values are used in the parametric study. These represent two characteristic deformations: one that exhibits clear back-relaxation behaviour and one that does not. The parameter sets were also adapted from [17], where they were fitted to different experimental results showing two distinct actuation profiles. These profiles were obtained by varying the water concentration in the samples, which was controlled by changing the number of chemical plating cycles during fabrication. In the present work, these experimentally calibrated parameters serve as two contrasting baselines, ensuring that the trends observed in the parametric study can be attributed to the effect of the individual parameter being varied, rather than to the specific combination of properties in a single baseline.

Table 1 shows the base parameters used for both cases, and table 2 shows the other parameters used in the simulation. The parameters in table 1 serve as the reference points for the parametric study, in which each parameter is varied individually while the others are held constant.

Table 1. Base parameters used for Case 1 (with back-relaxation) and Case 2 (without back-relaxation) [17].

Parameters	Case 1 (BR)	Case 2 (Without BR)
Dielectric constant	8×10^{8}	2.35×10^{9}
Ionic diffusivity ($m^2\,s^{-1}$)	1.4×10^{-11}	4×10^{-12}
Hydraulic permeability ($m^3 \cdot s\,kg^{-1}$)	1.52×10^{-17}	6.1×10^{-18}
Osmotic coefficient	1	3
Elastic modulus (Pa)	1.6×10^{9}	1.12×10^{9}

Table 2. Other constants used in the simulation [17].

Parameter	Value	Description
d_{ww}	$8 \times 10^{-11}\ m^2\,s^{-1}$	Diffusion coefficient of water
n_{dw}	4	Static drag coefficient of water
W_{v0}	0.5368	Initial volume fraction of water
c^{-}	1393	Initial concentration of fixed anions

Each of these parameters influences specific parts of the multiphysics model. The dielectric constant appears in the effective permittivity term in Poisson's equation (equation (1)), which governs the electric potential distribution. Changes in dielectric constant alter the electric field strength, thus affecting cation migration. The diffusion coefficient appears in the Nernst–Planck flux equations (equations (2) and (3)), determining the rate at which cations and water molecules migrate through the membrane. The hydraulic permeability, also in the flux equations, governs the movement of water in response to pressure gradients (hydrostatic and osmotic). Finally, the elastic modulus affects the hydrostatic pressure via the elastic stress contribution in equations (8) and (9), influencing the total internal pressure that drives water redistribution and resulting deformation.

For the voltage study, the parameters listed under Case 1 are used and kept constant while the amplitude and frequency of the applied voltage are swept.

3. Results

3.1. Parametric analysis

3.1.1. Dielectric constant. Figure 3 illustrates the vertical displacement of the IPMC actuator under different dielectric constant values for two cases: with (a) and without back-relaxation (b). It is observed that a higher dielectric constant leads to a slower initial response, requiring slightly more time to reach maximum displacement. This behaviour can be explained by the reduction in the overall potential experienced in the membrane with a higher dielectric constant. Despite this slower response, higher dielectric constants ultimately result in greater peak and steady-state displacements.

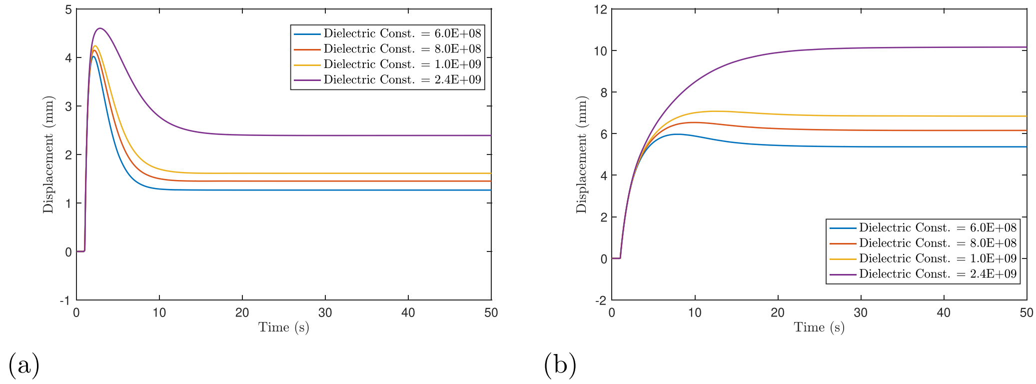

Figure 3. Displacement plot for dielectric constant sweep with base parameters that exhibits (a) back relaxation (case 1) and (b) no back relaxation (case 2).

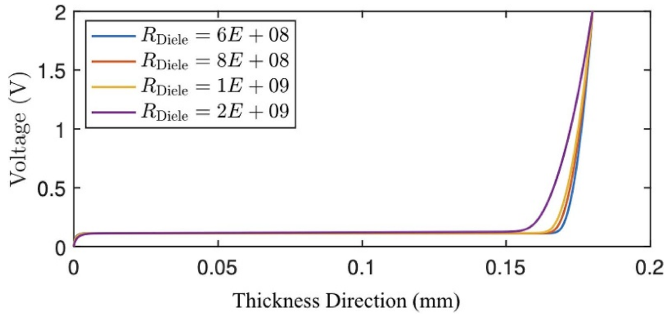

Figure 4. Voltage distribution across membrane thickness at steady-state for different dielectric constant.

It has been reported in the literature that the dielectric constant of Nafion membranes increases with water content [21]. Additionally, experimental studies have shown that IPMC deformation tends to increase with higher hydration levels to a certain extent [10, 22]. While water content also influences other material properties, these findings support the trend observed in our simulations: increasing the dielectric constant alone leads to greater deformation. This suggests that part of the experimentally observed enhancement in actuation with hydration can be attributed to the change in dielectric properties, validating the role of dielectric constant in influencing IPMC performance.

To further investigate the cause of the increase in the higher steady-state displacement, the cation concentration profile across the membrane thickness at steady-state is plotted. It can be seen in figure 4 that the cation depletion region is thicker with a higher dielectric constant. The thicker cation depletion region leads to greater contraction at the anode side, which creates a greater internal stress gradient, resulting in greater deformation.

In the case with back-relaxation (figure 3(a)), the dielectric constant shows a clear effect on both the transient and steady-state responses. However, in the case without back-relaxation (figure 3(b)), the influence of the dielectric constant on steady-state displacement becomes even more apparent. As the dielectric constant decreases, the steady-state displacement drops more significantly. The stronger impact on the steady-state displacement may contribute to the appearance of the back-relaxation phenomenon. These observations suggest that dielectric properties play a role in both the development and magnitude of relaxation behaviour.

3.1.2. Diffusion coefficient. Figure 5 shows the tip displacement of the IPMC actuator under varying diffusion coefficients, comparing the behaviour in cases with (a) and without (b) back-relaxation. The results indicate that the diffusion coefficient primarily affects the transient response, particularly the peak displacement, but has minimal influence on the final steady-state deformation. This trend can be attributed to the role of diffusion in controlling the rate at which cations and associated water molecules migrate through the polymer matrix under an applied electric field. In both scenarios, higher diffusion coefficients lead to faster ionic migration, enabling a more rapid accumulation of charge near the cathode and a more pronounced initial swelling. As a result, actuators with larger diffusion coefficients exhibit larger peak displacements. This effect is particularly evident in the case with no back-relaxation (figure 5(b)), where the actuator with the highest diffusion coefficient reaches the maximum displacement the fastest and with the highest amplitude, followed by a sharp decline due to redistribution. In contrast, lower diffusion values result in slower swelling, producing smaller peak responses and more gradual relaxation behaviour. Another notable observation is that for both cases, it is observed that the diffusion coefficient has an impact on the back-relaxation

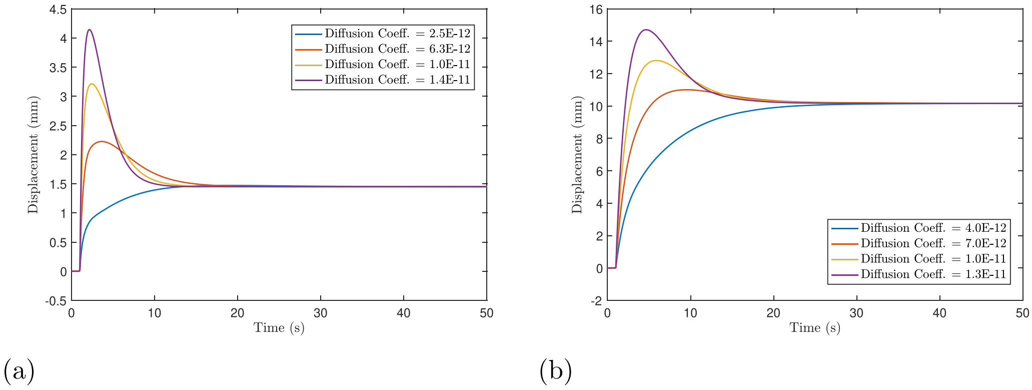

Figure 5. Displacement plot for diffusion coefficient sweep with base parameters that exhibits (a) back relaxation (case 1) and (b) no back relaxation (case 2).

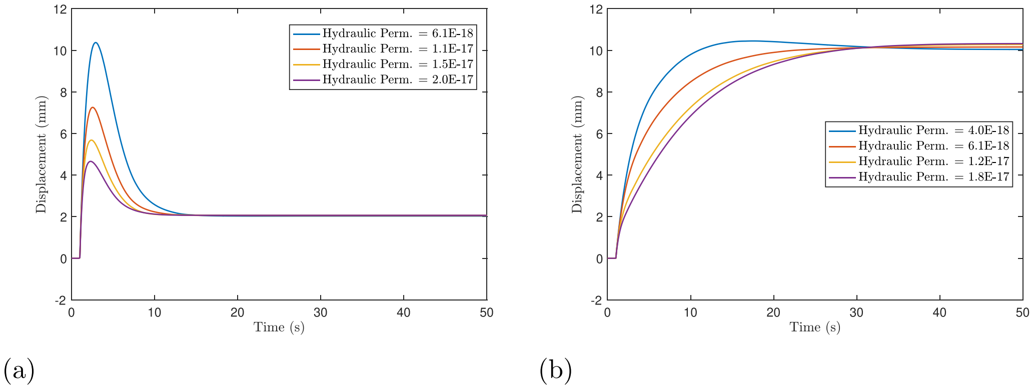

Figure 6. Displacement plot for hydraulic permeability sweep with base parameters that exhibits (a) back relaxation (case 1) and (b) no back relaxation (case 2).

behaviour. For instance, in the case with base parameters that show no back-relaxation as the diffusion coefficient increases, the back-relaxation becomes more pronounced.

This observation is consistent with experimental studies showing that the size and type of counterion can significantly affect actuation dynamics. Larger cations exhibit lower mobility in the polymer matrix due to steric hindrance between the channels, leading to reduced diffusion rates and slower deformation. As a result, IPMCs exchanged with larger cations can show smaller peak displacements, longer response times, and less relaxation compared to those with smaller cations [23]. Overall, the result suggests that the diffusion coefficient strongly influences the transient performance of IPMC actuators, determining the magnitude and how quickly the actuator reaches peak displacement, whereas its impact on steady-state displacement is negligible. This further emphasises that the diffusion coefficient governs the kinetics of deformation rather than the equilibrium state.

3.1.3. Hydraulic permeability. Figure 6 illustrates the influence of hydraulic permeability on the time-dependent displacement of the IPMC actuator. Similar to the diffusion coefficient, hydraulic permeability primarily governs the transient response of the actuator. Interestingly, the result suggests that lower hydraulic permeability leads to higher displacement.

Similar to the diffusion coefficient, cation size has also been reported to influence hydraulic permeability, where larger cations typically lead to reduced permeability [24]. As explained in the previous section, experimental studies suggest that larger cation size, therefore lower permeability, can result in lower actuation [23]. This may appear inconsistent with the simulation results presented here, where lower hydraulic permeability leads to greater peak displacement. However, this apparent contradiction can be explained by the way hydraulic permeability affects the convection-driven water flux in the model. In equations (2) and (3), we can see that the hydraulic permeability affects the convection flux, where the total water pressure consists of hydrostatic pressure and osmotic pressure. While the osmotic pressure is found to reduce the gradient of the total pressure across the membrane thickness, the sum of the two pressures still drives the water molecules/cations back to the anode as cations accumulate at the cathode. By increasing the hydraulic permeability, it amplifies the back flow of the cation and water, causing less swelling at the cathode, leading to a smaller displacement.

In the back-relaxation case (figure 6(a)), this effect is particularly evident: the actuator with the lowest permeability

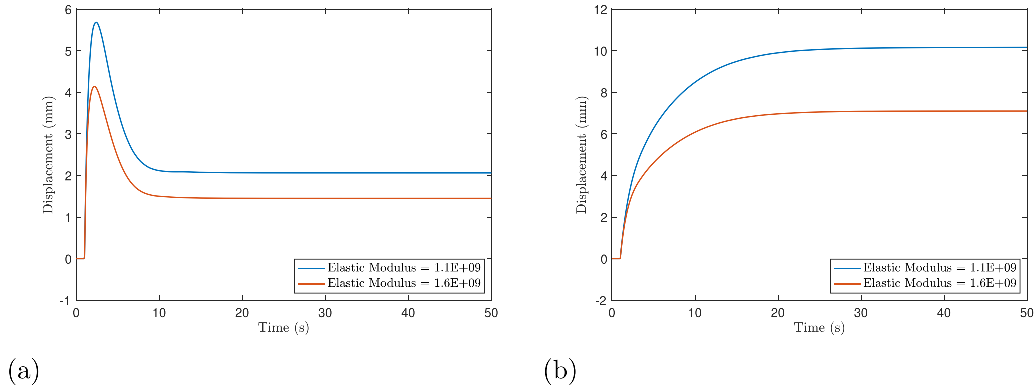

Figure 7. Displacement plot for elastic modulus sweep with base parameters that exhibits (a) back relaxation (case 1) and (b) no back relaxation (case 2).

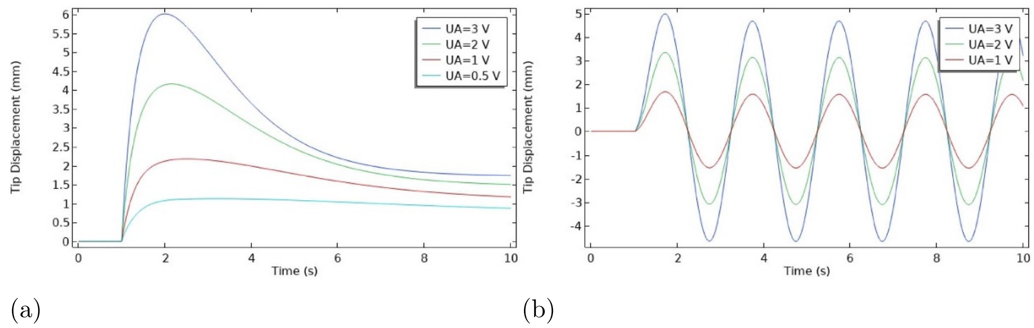

Figure 8. Displacement under different applied voltage for (a) step voltage and (b) sinusoidal voltage.

reaches a significantly larger peak before undergoing rapid back-relaxation, while lower permeability values result in more damped, slower deformation with lower peaks. Despite the pronounced differences in transient behaviour, all curves converge toward a similar steady-state displacement, indicating that it mainly influences the rate and extent of initial water redistribution but not the final deformation state.

The case without back-relaxation (figure 6(b)) again highlights its role in facilitating dynamic water movement. It is important to point out that the hydraulic permeability can have an impact on causing the back relaxation phenomenon, as it can be observed that when the parameter reaches a low enough value, the peak displacement becomes more prominent, therefore amplifying the back-relaxation phenomenon. However, the effect on back-relaxation from the hydraulic permeability is not as significant as the diffusion coefficient.

3.1.4. Elastic modulus. Figure 7 shows the effect of elastic modulus on the displacement response of IPMC actuators. In both cases, a lower elastic modulus results in larger overall deformation, consistent with the increased compliance of the material. This influence is evident not only in the peak displacement during the transient phase but also in the steady-state response, indicating that elastic modulus consistently affects the overall deformation dynamics. These results highlight that stiffness governs the actuator's ability to deform under electrochemical loading, impacting both the initial and sustained bending behaviour. This trend can be partially validated by experimental observations in the literature. For instance, it was reported that increasing the cross-link density, and thus the elastic modulus, of a fluorinated acrylic IPMC led to a reduction in actuation displacement [25]. Similarly, Shoji *et al* demonstrated that increasing relative humidity softens the membrane and enhances deformation [26]. While these studies also involve changes in ionic mobility and transport properties, the consistent inverse relationship between stiffness and displacement supports the role of elastic modulus as a key factor influencing actuation performance.

3.2. Voltage study

3.2.1. Amplitude and frequency sweep. Figure 8 shows the effect of voltage amplitude on the actuation behaviour of IPMCs under both DC and AC excitation. As expected, a higher input voltage leads to greater deformation in both cases. This trend is consistent with the underlying physics: a larger electric field enhances cation migration and water redistribution, resulting in more pronounced swelling near the cathode and, consequently, greater bending displacement.

With the DC input voltage, it can be observed more clearly that the voltage amplitude has more impact on the transient response. Higher voltages yield faster and larger

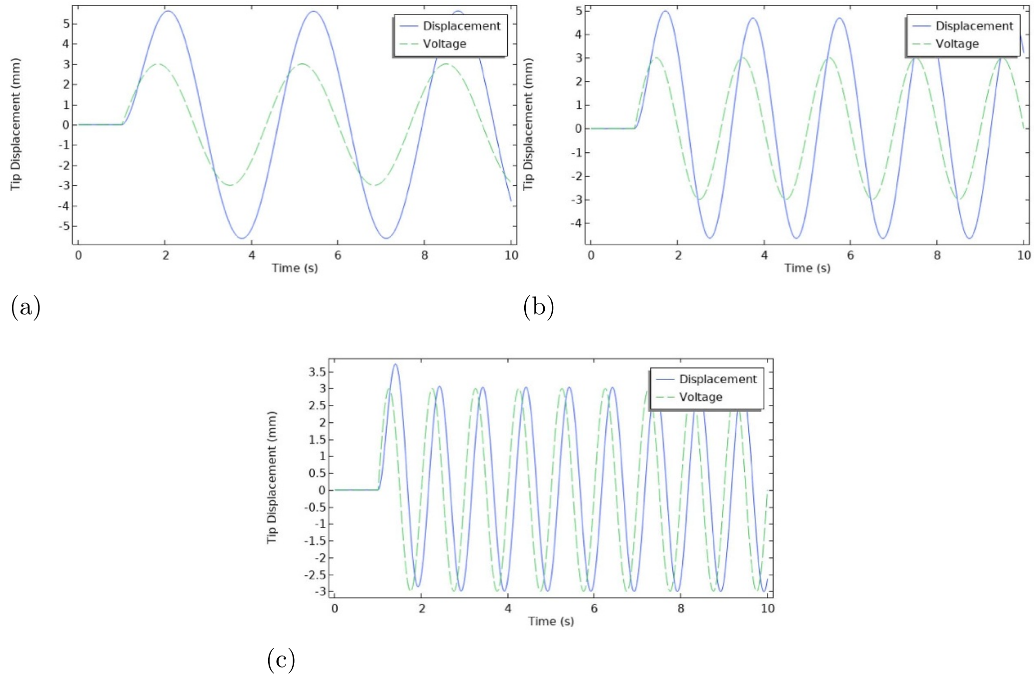

Figure 9. Displacement (solid line) and voltage (dashed line) under AC input frequency of: (a) 0.3 Hz, (b) 0.5 Hz, and (c) 1 Hz.

initial deformation, followed by a pronounced back-relaxation effect. Conversely, lower voltage inputs produce more damped responses with smaller initial displacement and noticeably weaker back-relaxation. This observation supports findings in previous studies [27, 28], which report that reduced driving voltages limit over-accumulation of ions near the electrodes and thus suppress the internal pressure imbalances that drive relaxation. The results suggest that lowering the DC input can be an effective strategy for improving long-term stability, albeit at the cost of actuation amplitude.

Under sinusoidal AC loading (figure 8(b)), the deformation amplitude scales proportionally with voltage, consistent with linear actuation behaviour at low frequencies. The simulated result agrees with many experimental results presented in literature [29, 30]. Besides investigating the effect of voltage amplitude, figure 9 shows how different frequencies of the AC voltage can also affect the level of deformation, where a higher frequency can lead to smaller deformation. With a higher frequency, the cations are not given enough time to fully migrate to the cathode therefore smaller deformation. This simulated result aligns with what is observed in literature [29, 31, 32].

3.2.2. AC load with bias. Building on the observed effects of DC and AC voltage inputs, we investigate whether combining a lower DC bias with a higher AC amplitude can maintain peak actuation performance while reducing the back-relaxation commonly associated with higher DC loads. This approach is motivated by earlier observations that lower DC voltages lead to less relaxation but also lower peak displacement. By introducing a high-amplitude AC signal, we aim to recover the lost displacement while leveraging the low-relaxation benefits of the lower bias.

Figure 10 shows the time-dependent displacement of the IPMC actuator under two different biased AC voltage conditions:

- Case 1 (LB-HA): 1 V DC bias with 2 V AC amplitude
- Case 2 (HB-LA): 2 V DC bias with 1 V AC amplitude

Both cases apply the same combined voltage magnitude (3 V total), but differ in how the AC and DC components are balanced. The result shows that LB-HA achieves a comparable maximum displacement to HB-LA, while exhibiting significantly reduced back-relaxation. This demonstrates that the low-bias, high-amplitude strategy can preserve strong actuation performance while improving long-term deformation stability.

This result supports our hypothesis and suggests that biased AC loading with a carefully selected balance of DC and AC components can be an effective strategy to control both the magnitude and stability of IPMC actuation. The high-frequency AC component likely helps maintain ionic activity and delays the redistribution processes responsible for relaxation, while the reduced DC bias minimises over-accumulation of ions near the electrodes.

The results presented above for both the parametric and voltage study do not account for water electrolysis or evaporation effects, which can become more significant at higher applied voltages. Therefore, the findings are most representative of submerged or high-humidity conditions, where membrane hydration can be reasonably assumed to remain constant. When the high voltage amplitudes are applied in air, two additional processes become important: (i) evaporative water loss and (ii) water electrolysis that occurs at higher voltage. Both processes can reduce the membrane's water content, which affects the material parameters such as

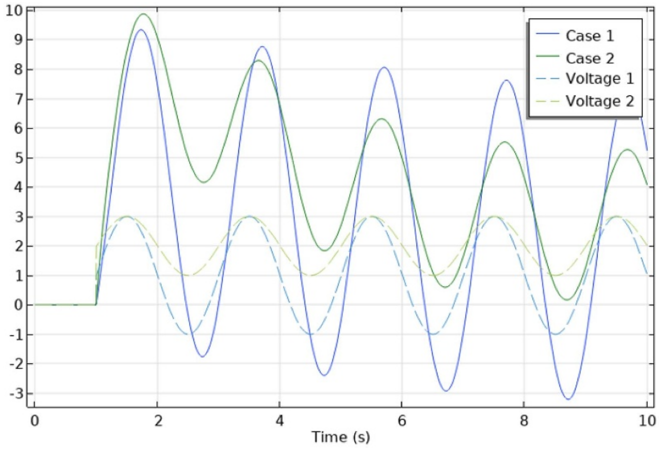

Figure 10. Tip displacement under two biased AC voltage inputs: 2 V AC with 1 V bias (low-bias, high-AC) and 1 V AC with 2 V bias (high-bias, low-AC).

diffusion coefficient and hydraulic permeability over time. Consequently, high-voltage experiments performed in air are expected to deviate from the trends reported here, not because the underlying transport-mechanical coupling is different, but because the governing parameters change as the membrane dehydrates. Capturing that regime would require extending the current framework to include water-content evolution and electrolysis kinetics. Nonetheless, both studies remain meaningful for understanding how different parameters influence the actuator's performance under stable hydration conditions. Since all simulations are performed under consistent initial conditions, the observed trends highlight the system's sensitivity to each parameter and provide useful insight into how these parameters influence the overall deformation dynamics.

4. Conclusion and future work

This study presents a multiphysics simulation-based parametric analysis to understand how intrinsic material parameters and electrical loading conditions influence the actuation behaviour of IPMC actuators. Using a finite element model adapted from Zhu *et al,* we separately investigated the effects of dielectric constant, ionic diffusivity, hydraulic permeability, and elastic modulus under step voltage input, along with the impact of DC and AC voltage parameters. The findings from this study are listed below:

- Parameters controlling transport dynamics, such as diffusivity and permeability, primarily affect the transient response, modulating how fast and how much the actuator bends initially, while having minimal influence on steady-state displacement.
- Parameters that influence the internal charge and water distribution, such as dielectric constant and elastic modulus, significantly impact both the peak and steady-state deformation.
- Transport parameters can suppress or trigger back-relaxation, depending on how they alter pressure buildup and ion migration over time.
- The voltage study demonstrated that combining a high AC amplitude with a low DC bias can maintain high peak displacement while mitigating back-relaxation, offering a promising control strategy for improving long-term actuator stability.

These findings not only enhance our understanding of the underlying physics in IPMC deformation but also offer practical guidance for tuning material and operational parameters to meet application-specific performance goals. Future research could expand on the current study by:

- Explore the impact of varying key parameters across different physical models proposed in the literature to assess whether consistent trends in deformation behaviour can be observed.
- Implementing the complete eigenstress formulation, allowing a more detailed investigation into how various parameters independently influence the osmotic and electrostatic stress components.
- Conduct experiments to further validate the simulation findings and evaluate the practical implementation feasibility.
- The effect of frequency in biased AC input should be explored, as higher frequencies can limit the duration available for water electrolysis to occur. Preliminary result also suggests that implementing higher frequencies may help suppress back-relaxation by introducing disturbances that hinder free hydrated cations from migrating back toward the anode [33]. However, further investigation is needed to identify optimal combinations of AC amplitude, DC bias, and frequency that maximise actuation performance while maintaining stability.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Author contributions

Yu-Tung Chen 0009-0002-6706-4466
Conceptualization (lead), Data curation (lead), Formal analysis (lead), Investigation (lead), Methodology (lead), Software (lead), Validation (lead), Visualization (lead), Writing – original draft (lead), Writing – review & editing (equal)

Kamran Behdinan 0000-0002-1873-9837
Funding acquisition (lead), Project administration (lead), Resources (lead), Supervision (lead), Writing – review & editing (equal)

References

[1] He Q, Yin G, Vokoun D, Shen Q, Lu J, Liu X, Xu X, Yu M and Dai Z 2022 Review on improvement, modeling and application of ionic polymer metal composite artificial muscle *J. Bionic Eng.* **19** 279–98

[2] Yu M, Shen H and Dai Z 2007 Manufacture and performance of ionic polymer-metal composites *J. Bionic Eng.* **4** 143–9

[3] MohdIsa W, Hunt A and HosseinNia S 2019 Sensing and self-sensing actuation methods for ionic polymer-metal composite (IPMC): a review *Sensors* **19** 3967

[4] Bar-Cohen Y *et al* 2001 Electroactive polymer actuators as artificial muscles *Proc. SPIE (Washington)*

[5] Jung K, Nam J and Choi H 2003 Investigations on actuation characteristics of IPMC artificial muscle actuator *Sens. Actuators A* **107** 183–92

[6] Hao M, Wang Y, Zhu Z, He Q, Zhu D and Luo M 2019 A compact review of IPMC as soft actuator and sensor: current trends, challenges and potential solutions from our recent work *Front. Robot. AI* **6** 129

[7] Saccardo M, Zuquello A, Tozzi K, Gonçalves R, Hirano L and Scuracchio C 2020 Counter-ion and humidity effects on electromechanical properties of Nafion®/Pt composites *Mater. Chem. Phys.* **244** 122674

[8] Shahinpoor M and Kim K 2000 Effects of counter-ions on the performance of IPMCs *Proc. SPIE* **3987** 110–20

[9] Onishi K, Sewa S, Asaka K, Fujiwara N and Oguro K 2001 The effects of counter ions on characterization and performance of a solid polymer electrolyte actuator *Electrochim. Acta.* **46** 1233–41

[10] Zhu Z, Chang L, Asaka K, Wang Y, Chen H, Zhao H and Li D 2014 Comparative experimental investigation on the actuation mechanisms of ionic polymer-metal composites with different backbones and water contents *J. Appl. Phys.* **115** 124903

[11] Lee S, Han M, Kim S, Jho J, Lee H and Kim Y 2006 A new fabrication method for IPMC actuators and application to artificial fingers *Smart Mater. Struct.* **15** 1217

[12] Oh C, Kim S, Kim H, Park G, Kim J, Ryu J, Li P, Lee S, No K and Hong S 2019 Effects of membrane thickness on the performance of ionic polymer-metal composite actuators *RSC Adv.* **9** 14621–6

[13] Li S and Yip J 2019 Characterization and actuation of ionic polymer metal composites with various thicknesses and lengths *Polymers* **11** 91

[14] Dhar Badgayan N, Kumar Sahu S, Samanta S and Rama Sreekanth P 2018 Assessment of bulk mechanical properties of HDPE hybrid composite filled with 1D/2D nanofiller system *Mater. Sci. Forum.* **917** 12–16

[15] Amirapu S, Nelapati G, Yalamanchili H, Badgayan N and Sahu S 2022 HDPE based polymeric nanodiamond nanocomposite for total knee arthoplasty: a finite element based approach *Mater. Today: Proc.* **56** 1622–8

[16] Yesaswi C, Sahu S and Sreekanth P 2022 Experimental investigation of electro-mechanical behavior of silver-coated teflon fabric-reinforced nafion ionic polymer metal composite with carbon nanotubes and graphene nanoparticles *Polymers* **14** 5497

[17] Zhu Z, Wang Y, Liu Y, Asaka K, Sun X, Chang L and Lu P 2016 Application-oriented simplification of actuation mechanism and physical model for ionic polymer-metal composites *J. Appl. Phys.* **120** 034901

[18] Porfiri M 2008 Charge dynamics in ionic polymer metal composites *J. Appl. Phys.* **104** 104915

[19] Pugal D, Kim K, Punning A, Kasemägi H, Kruusmaa M and Aabloo A 2008 A self-oscillating ionic polymer-metal composite bending actuator *J. Appl. Phys.* **103** 084908

[20] Vokoun D, He Q, Heller L, Yu M and Dai Z 2015 Modeling of IPMC cantilever's displacements and blocking forces *J. Bionic Eng.* **12** 142–51

[21] Paddison S, Reagor D and Zawodzinski T 1998 High frequency dielectric studies of hydrated Nafion® *J. Electroanal. Chem.* **459** 91–97

[22] Yeh C and Shih W 2010 Effects of water content on the actuation performance of ionic polymer-metalcomposites *Smart Mater. Struct.* **19** 124007

[23] Nemat-Nasser S and Wu Y 2006 Tailoring the actuation of ionic polymer-metal composites *Smart Mater. Struct.* **15** 909

[24] Evans C, Noble R, Nazeri-Thompson S, Nazeri B and Koval C 2006 Role of conditioning on water uptake and hydraulic permeability of Nafion® membranes *J. Membr. Sci.* **279** 521–8

[25] Mo Jeong H, Soo Kim H and Soo La Y 2006 The effect of cross-linking on the actuation of an electroactive IPMC prepared with a fluorinated acrylic copolymer *J. Macromol. Sci.* **45** 119–30

[26] Shoji E and Hirayama D 2007 Effects of humidity on the performance of ionic polymer- metal composite actuators: experimental study of the back-relaxation of actuators *J. Phys. Chem.* **111** 11915–20

[27] Porfiri M, Leronni A and Bardella L 2017 An alternative explanation of back-relaxation in ionic polymer metal composites *Extreme Mech. Lett.* **13** 78–83

[28] Del Bufalo G, Placidi L and Porfiri M 2008 A mixture theory framework for modeling the mechanical actuation of ionic polymer metalcomposites *Smart Mater. Struct.* **17** 045010

[29] Ma S, Zhang Y, Liang Y, Ren L, Tian W and Ren L 2020 High-performance ionic-polymer-metal composite: toward large-deformation fast-response artificial muscles *Adv. Funct. Mater.* **30** 1908508

[30] Yin G, He Q, Zhou X, Wu Y, Li H and Yu M 2021 Printing ionic polymer metal composite actuators by fused deposition modeling technology *Int. J. Smart Nano Mater.* **12** 218–31

[31] Palmre V, Kim S, Pugal D and Kim K 2014 Improving electromechanical output of IPMC by high surface area Pd-Pt electrodes and tailored ionomer membrane thickness *Int. J. Smart Nano Mater.* **5** 99–113

[32] Zhang T, Chen K and Wu J 2023 Deformation performance of Pt-IPMC based on experiment and Simulation Research *Integr. Ferroelectr.* **232** 19–28

[33] Annabestani M Sayad M Esmaeili-Dokht P and Fardmanesh M 2021 Toward a high performance IPMC soft actuator using a disturbance-aided method (arXiv:2106.13040)